The top two images are shortly before the beginning and shortly after the end of Figure 9.5; I enlarged these two images tenfold compared to the others, to make the hole and the disrupting star visible.

As the whole set of images shows, over the subsequent several years much of the star's matter is captured into orbit around the black hole, where it begins to form an accretion disk. And the remaining matter escapes from the hole's gravitational pull along a streaming, jetlike trajectory.

Gargantua's Accretion Disk and Missing Jet

A typical accretion disk and its jet emit radiation—X-rays, gamma rays, radio waves, and light—radiation so intense that it would fry any human nearby. To avoid frying, Christopher Nolan and Paul Franklin gave Gargantua an exceedingly anemic disk.

Now, "anemic" doesn't mean anemic by human standards; just by the standards of typical quasars. Instead of being a hundred million degrees like a typical quasar's disk, Gargantua's disk is only a few thousand degrees, like the Sun's surface, so it emits lots of light but little to no X-rays or gamma rays. With gas so cool, the atoms' thermal motions are too slow to puff the disk up much. The disk is thin and nearly confined to Gargantua's equatorial plane, with only a little puffing.

Disks like this might be common around black holes that have not torn a star apart in the past millions of years or more—that have not been "fed" in a long time. The magnetic field, originally confined by the disk's plasma, may have largely leaked away. And the jet, previously powered by the magnetic field, may have died. Such is Gargantua's disk: jetless and thin and relatively safe for humans. Relatively.